

Neural SC Decoder for MAC Polar Codes

Results across channels and code classes

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1. Setup

Two-user binary-input MAC with polar codes. Two code classes:

- **Class B:** path $0^{N/2}1^N 0^{N-i}$
- **Class C:** path $0^N 1^N$ (corner rate). Decomposes into two chained single-user decoders.

Neural decoder architectures:

- **NCG decoder:** CalcLeft/CalcRight/CalcParent, smooth logits2emb, 4-class joint (u, v) output, sequential training. Used for Class B.
- **Chained NPD:** two single-user NPDs (CheckNode/BitNode, sign-flip residual, binary output, fast_ce parallel training). Used for Class C.

2. Channel capacities

Channel	$I(X; Z)$	$I(Y; Z X)$	$I(X, Y; Z)$	Notes
GMAC (SNR=6dB)	0.465	0.912	1.376	Gaussian noise, BPSK
BEMAC	0.500	1.000	1.500	Deterministic $Z = X + Y$
ABNMAC	0.400	0.800	1.200	Correlated binary noise

3. GMAC (Gaussian MAC, SNR = 6 dB)

3.1 Class B

NCG neural decoder matches SC up to $N = 128$.

N	k_u	k_v	SC BLER	NCG BLER	Ratio
16	8	8	0.070		
32	15	15	0.047	0.040	0.87x
64	31	31	0.028	0.026	1.03x
128	62	62	0.020	0.023	1.43x
256	123	123	0.006		
512	246	246	0.002		
1024	492	492	0.002		

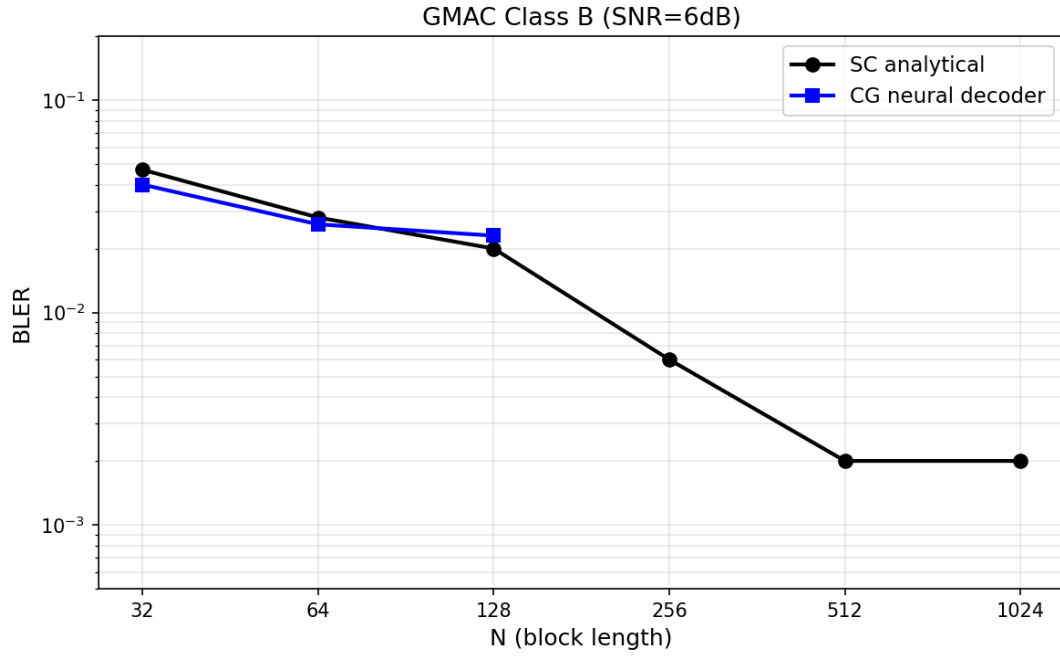


Figure 1: GMAC Class B: SC analytical and NCG neural decoder.

3.2 Class C

Chained NPD beats SC at all tested N .

N	k_u	k_v	SC BLER	NPD BLER	Ratio	CW
16	4	7	0.163	0.111	0.68x	2000
32	7	15	0.068	0.038	0.56x	2000
64	15	29	0.027	0.017	0.63x	2000
128	30	58	0.005	0.007	1.40x	2000
256	59	117	0.002	0.0016	0.80x	5000
512	119	233	0.0008			
1024	238	467	0.0002			

$N = 128$ NPD result uses the $N = 256$ trained model (weight-shared architecture generalizes across N).

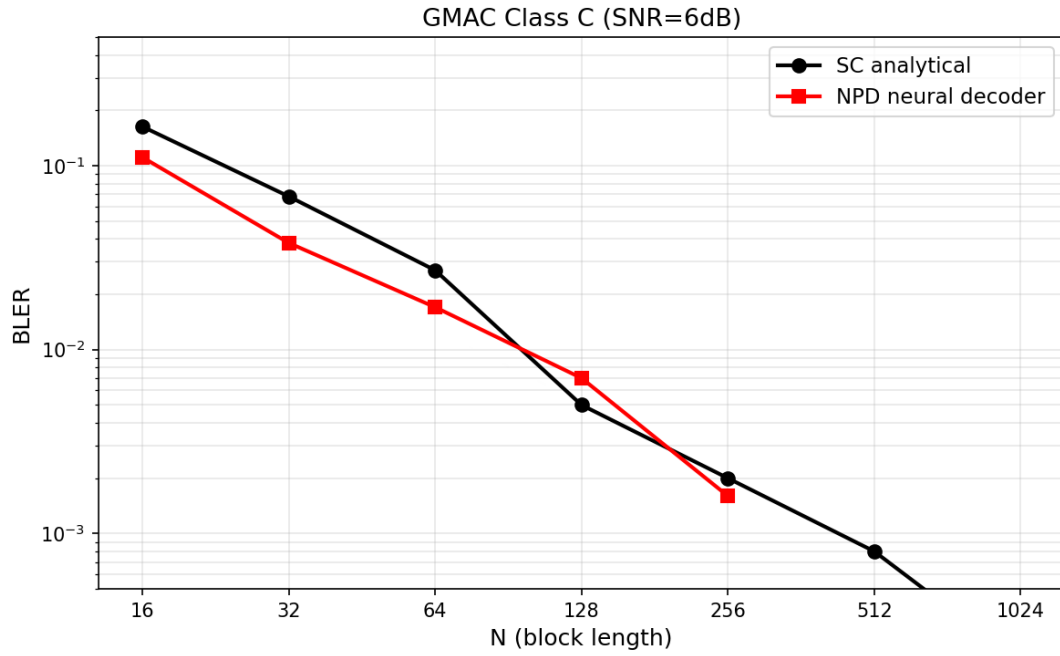


Figure 2: GMAC Class C: SC analytical and chained NPD.

4. BEMAC (Binary Erasure MAC)

4.1 Class B

NCG neural decoder matches or beats SC at all N .

N	SC BLER	NCG BLER	Ratio	CW
16	0.011	0.011	1.08x	5000
32	0.008	0.009	1.10x	5000
64	0.006	0.003	0.54x	5000
128	0.002	0.001	0.60x	5000
256	8e-5	4e-5	0.50x	50000
1024	1e-4	1e-4	1.0x	10000

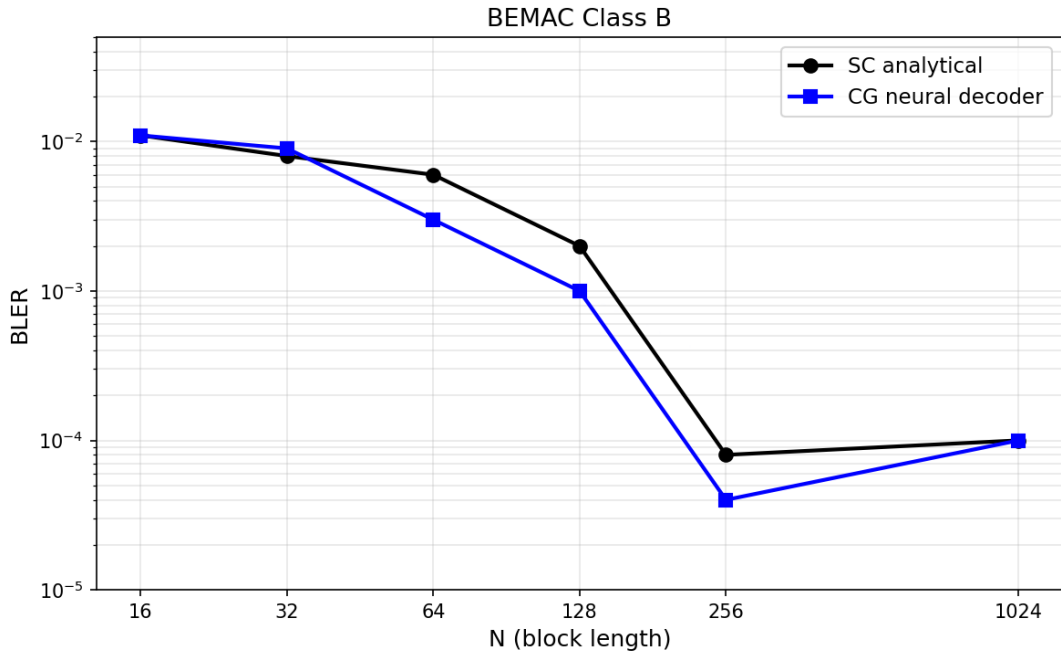


Figure 3: BEMAC Class B: SC analytical and NCG neural decoder.

4.2 Class C

SC analytical baselines (neural decoder pending).

N	k_u	k_v	SC BLER
16	4	7	0.163
32	7	15	0.068
64	15	29	0.027
128	30	58	0.005
256	59	117	0.002

5. ABNMAC (Asymmetric Binary Noise MAC)

SC analytical baselines (neural decoder pending).

5.1 Class B

N	k_u	k_v	SC BLER
16	3	6	0.078
32	6	13	0.125
64	13	26	0.092
128	26	51	0.071
256	51	102	0.043

5.2 Class C

N	k_u	k_v	SC BLER
16	3	6	0.062
32	6	13	0.025
64	13	26	0.050
128	26	51	0.028
256	51	102	0.013

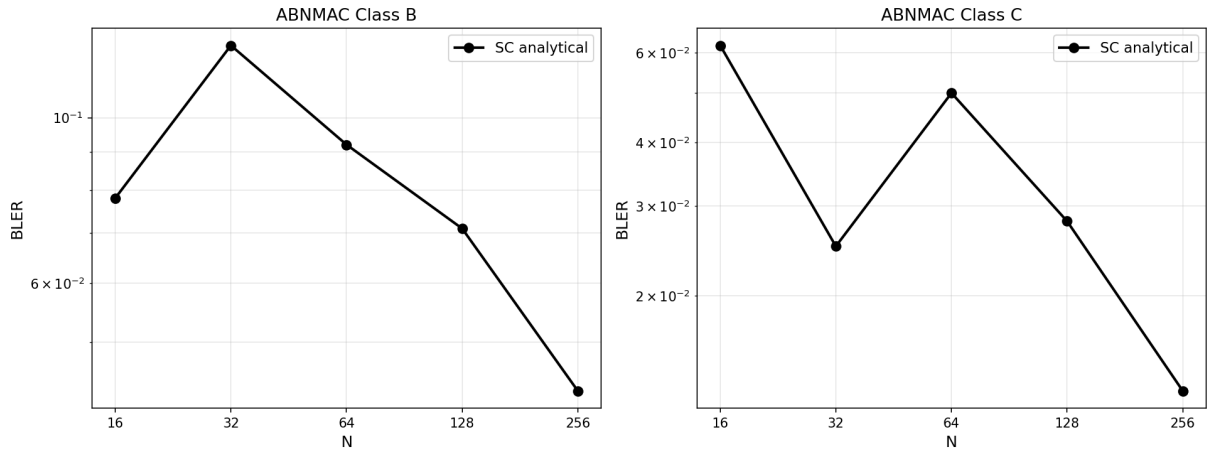


Figure 4: ABNMAC: SC analytical baselines for Class B (left) and Class C (right).

6. Key insight: NPD-guided code design

The Class C results were enabled by jointly designing the polar code with the neural decoder. Selecting information positions by per-position MI from the trained NPD instead of genie-aided SC design:

Decoder	Frozen set	$N = 32$ BLER
SC	SC-optimal (genie)	0.068
NPD	SC-optimal	0.130
NPD	NPD-optimal	0.034

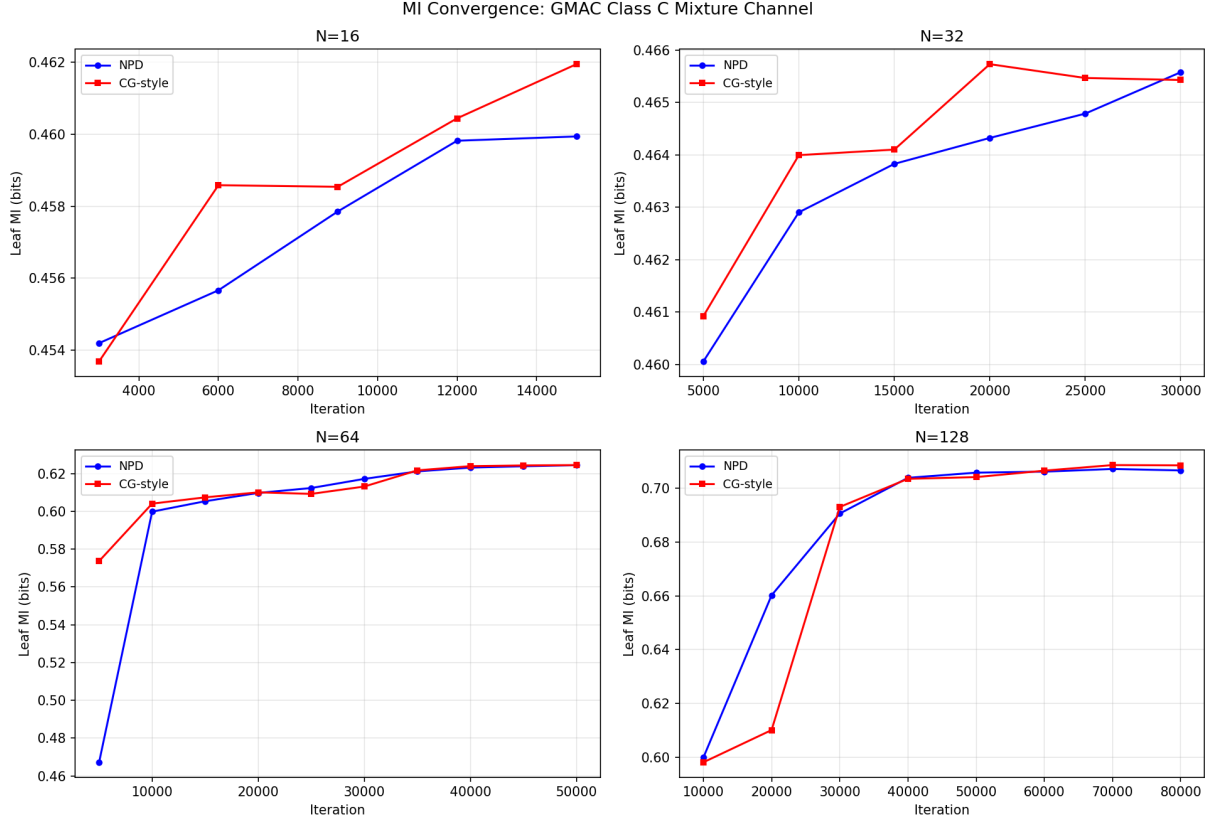


Figure 5: MI convergence during training: NPD and NCG-style at $N = 16, 32, 64, 128$.

7. Summary

Channel	Class	Neural decoder	Best N range	vs SC
GMAC	B	NCG	$N = 32\text{--}64$	0.87–1.03x
GMAC	C	NPD	$N = 16\text{--}256$	0.56–0.80x
BEMAC	B	NCG	$N = 16\text{--}1024$	0.50–1.10x
BEMAC	C	–		pending
ABNMAC	B	–		pending
ABNMAC	C	–		pending

8. TODO

Neural decoder training needed:

- ABNMAC Class B: NCG decoder
- ABNMAC Class C: chained NPD
- BEMAC Class C: chained NPD

Scaling:

- GMAC Class B NCG: close the gap at $N = 256$ (currently 3.4x SC)
- GMAC Class C NPD: verify $N = 512$, $N = 1024$

Methodology:

- Apply NPD-guided code design to Class B
- Apply NPD training methodology (rate-1 training, MI tracking) to the NCG decoder
- Extend to channels with memory (ISI-MAC)